

Introduction to Electronic & Electrical Technology

Engineering Technology

May, 2026



Introduction to Electronic & Electrical Technology (A25223)

Short Title:	Intr to E & E Tech	
Faculty	Science and Computing	
Department	Engineering Technology	
Cluster	Engineering	
Author	CLAIRE.FITZPATRICK	
Credits	5	Level: Introductory

Description of Module / Aims

The aim of this module is to introduce students to the fundamentals of Digital Electronics and also to introduce the students to basic Electrical Installation practice and Electrical Safety. The principles of combinational and sequential logic, circuit elements and their applications are covered. Essential electrical safety, protection circuit elements, and good wiring circuitry implementation are included via a comprehensive workshop based element. Students will also use CAD to produce electrical drawings.

Programmes

BEng (Hons) in Automation Engineering with Data Intelligence (WD_EAUTO_B)	1 / 2 / E
BEng (Hons) in Electrical Engineering (WD_ETRIC_B)	1 / 2 / E
BEng (Hons) in Electrical and Automation Engineering (International) (WD_ETRIC_BI)	1 / 1 / M
BEng (Hons) in Electronic Engineering (WD_EONIC_B)	1 / 2 / E
BEng (Hons) in Information Engineering (International) (WD_EEELC_BI)	1 / 1 / M
BEng (Hons) (WD_ETRIC_B)	1 / 2 / E

Indicative Content

- Applications of Boolean Algebra
- Combinational and sequential logic circuits and applications
- Arithmetic circuits, oneshots, multiplexers, decoders and encoders
- Electrical and personal safety, Earthing systems, fuses, MCB's and RCB's
- Cable installation and termination techniques, Electrical Wiring systems
- Electrical Drawing and schematic generation.

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Explain the fundamental concepts of combinational and sequential logic design.
2. Evaluate, simplify and convert Boolean expressions.
3. Define, analyse, design and build combinational and sequential logic circuits using SSI/MSI components.
4. Be aware of electrical safety procedures and the function of earthing systems, fuses, MCB's and RCB's within Electrical circuits.
5. Demonstrate an ability to construct simple electrical wiring systems and basic electrical control circuits.
6. Generate an electrical schematic using appropriate CAD tools,.

Learning and Teaching Methods

- Lectures, presentations, reading and problem solving in class.
- Practical electrical installations will be performed in the Workshop
- The student will read supplementary material in their own time.
- Project work, as assigned by the lecturer.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	1,2,3,4,5,6
<i>Practical</i>	35%	4,5
<i>Portfolio</i>	15%	6
<i>In-Class Assessment</i>	30%	1,2
<i>Lab Report</i>	20%	3

Assessment Criteria

70% -

100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Essential Material

- Floyd, T.L. *Digital Fundamentals*. 11th Ed. U.S.: Pearson Education, 2014.
- Scaddan, B. *The Dictionary of Electrical Installation Work*. UK: Newnes, 2011.
- Scaddan, B. *Wiring Systems and Fault Finding*. UK: Newnes, 2008.

Supplementary Material

- Linsley, T. _Introduction to Electrical Installation Work_. USA: Elsevier Science & Technology, 2011.
- NSAI, Standard. _Good Practice Guide on the Management of Electricity at Work_. Ireland: NSAI, 2009.

Requested Resources

- Lecture Room: Loose Seated
- ENGINEERING LAB: Electronics
- Room Type: Computer Lab
- ENGINEERING WORKSHOP: Electrical Workshop